

River Kest Bridge Ground Investigation Report

Project	Kestrel Valley Link Road
Document	KVL-SGI-410-RP-T-0001
Revision	P01
Status	FI - For information
Date	18 October 2024
Originator	Stratum Ground Investigations

Rev	Date	Description
P01	18/10/2024	First issue

1 Introduction

Stratum Ground Investigations was appointed by Arden Moss Consulting Engineers to investigate ground conditions at the proposed River Kest Bridge on the Kestrel Valley Link Road. This report presents the factual records and our interpretation for the design of the abutment foundations.

2 Fieldwork

Five cable percussive boreholes with rotary follow-on were completed between 2 and 20 September 2024: BH-W1 and BH-W2 at the west abutment, and BH-E1, BH-E2 and BH-E3 at the east abutment. BH-E3 was added late at the request of the designer and was completed on 20 September 2024.

3 Ground conditions

The sequence comprises made ground over soft to firm alluvial clay, dense sand and gravel, and weathered mudstone. At BH-E1 and BH-E2 the mudstone was proven from about -4.5 mAOD.

The interpretation in this section is based on BH-W1 to BH-E2. The results of BH-E3 were received after the interpretive assessment was completed and are presented in Appendix C.

Borehole	Ground level (mAOD)	Top of mudstone (mAOD)	Notes
BH-W1	13.60	-3.90	
BH-W2	13.75	-4.10	
BH-E1	14.20	-4.40	
BH-E2	14.15	-4.60	
BH-E3	14.30	-9.80	See Appendix C

4 Recommendations

Bored cast-in-place piles founded in the weathered mudstone are recommended at both abutments. Pile lengths of the order of 18 m are expected at the east abutment.

Appendix C - BH-E3 (late borehole)

BH-E3 encountered a lens of very soft to soft grey clay between -2.1 mAOD and -6.4 mAOD, below the dense gravel, with mudstone not proven until -9.8 mAOD. The lens was not encountered at BH-E1 or BH-E2. Its lateral extent is unknown.

Prepared by: Emma Price, Principal Geotechnical Engineer, Stratum Ground Investigations